Solar Plane Design Report (Detailed, with MPPT + Graphs)

Project workspace: C:\Users\Janal\NewPWS

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1. Project Inputs and Constraints

- Airframe mass used in analysis: 1.851 kg (your 1.70 kg baseline plus servos, wiring, MPPT allowance).
- Wing geometry: span 3.00 m, area 0.600 m^2, aspect ratio 15.00.
- Solar cells: 21 cells of 125 mm x 125 mm mono.
- Battery bay hard limits: depth 54 mm, width 31 mm (+3 mm tolerated), height 19 mm.
- Existing propulsion hardware: D3530 1100KV, 14x4.7 slow-fly prop (CW), 50A ESC.

Chosen control/telemetry chain (user-selected)

- SpeedyBee F405
- UART MAVLink link to Raspberry Pi Zero 2 W
- mavlink-router on Pi
- A7670 LTE modem for WAN uplink
- Internet tunnel to QGroundControl
- Avionics power included in calculations: FC 1.0 W + Pi 1.8 W + LTE modem avg 3.0 W + misc 0.5 W = 6.3 W.
- Compatibility check: UART level is 3.3V on Pi GPIO UART and typical FC UART logic, so direct serial data interface is valid when grounds are shared.
- Risk 1 (power): A7670-class LTE modules have high TX current pulses; do not power the modem from Pi 5V pin. Use a dedicated regulator and local bulk capacitance.
- Risk 2 (control): cellular latency/coverage can drop unexpectedly; keep ELRS as primary C2 and use LTE for telemetry/backup control path.
- Risk 3 (firmware headroom): F405 targets can be flash-limited for advanced ArduPilot feature sets; verify your exact target build fits required features before finalizing.

2. Core Performance Results

- Estimated stall speed: 6.42 m/s
- Best-endurance speed: 8.25 m/s
- Mission speed used (1.4 x stall margin): 8.98 m/s
- Propulsion electrical power at best-endurance speed: 17.95 W
- Total electrical power at best-endurance speed (propulsion + avionics): 24.25 W
- Total electrical power at mission speed: 27.71 W (propulsion 21.41 W + avionics 6.30 W)
- Motor estimate: loaded RPM 5417 rpm, current 28.0 A, static thrust 12.0 N

3. Flight Power and Duration Graphs

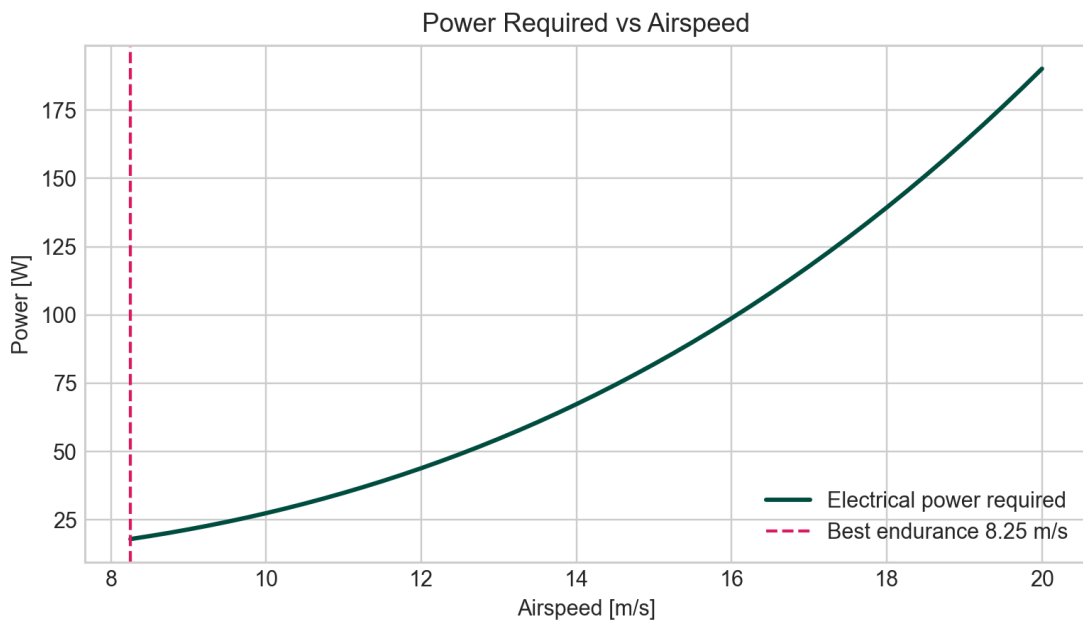

Figure 1: Propulsion electrical power required vs airspeed.

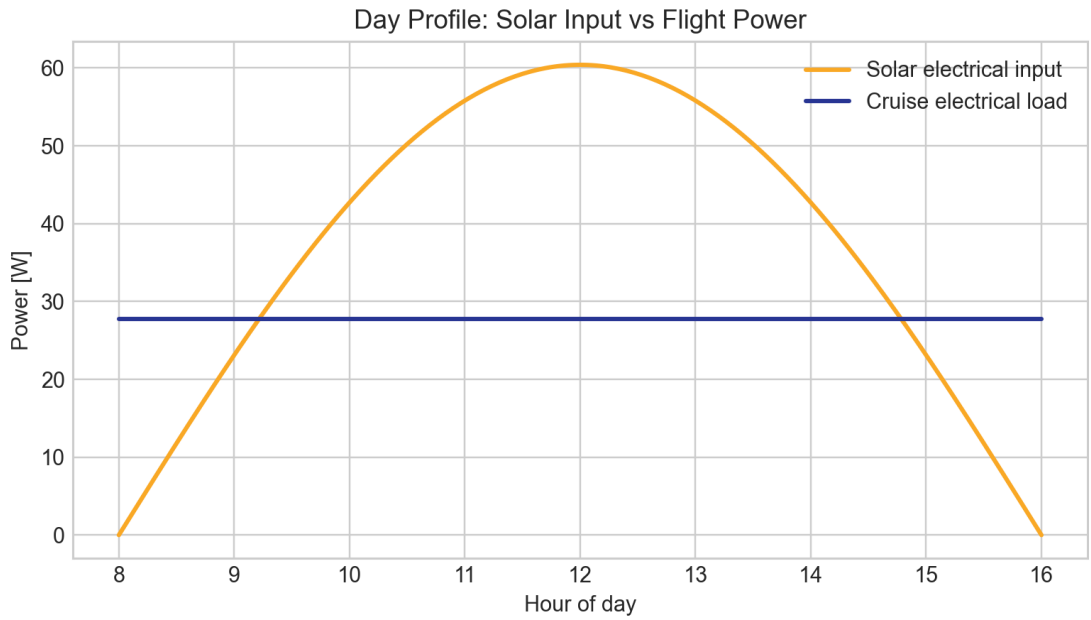

Figure 2: Day profile power balance: solar input vs total cruise load (propulsion + avionics).

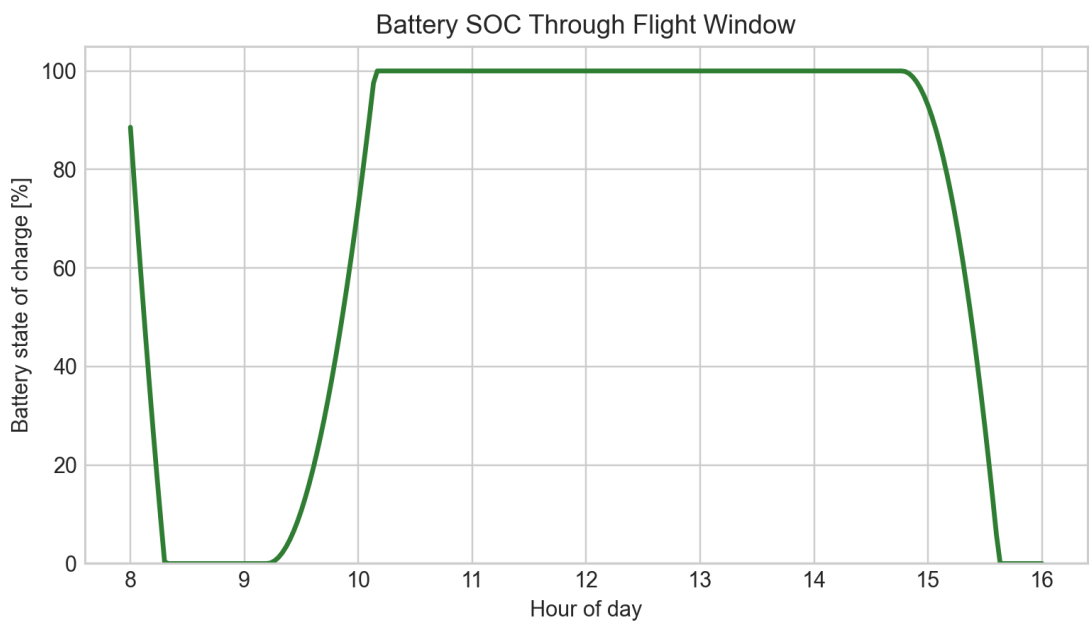

Figure 3: Battery state of charge through the daylight mission window.

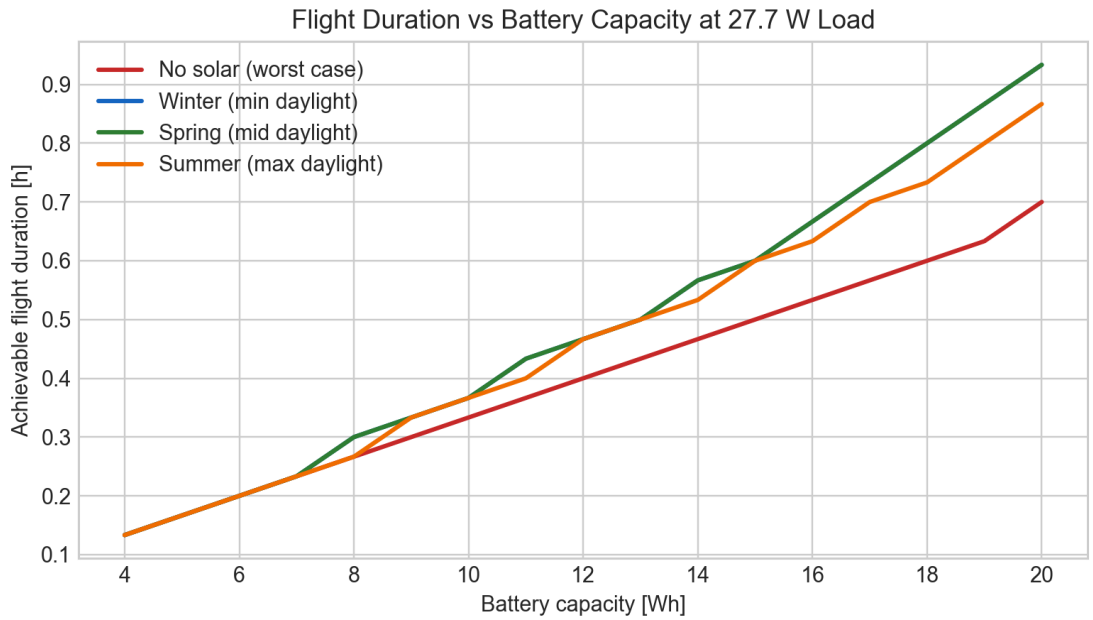

Figure 4: Flight duration versus battery capacity (with and without solar support).

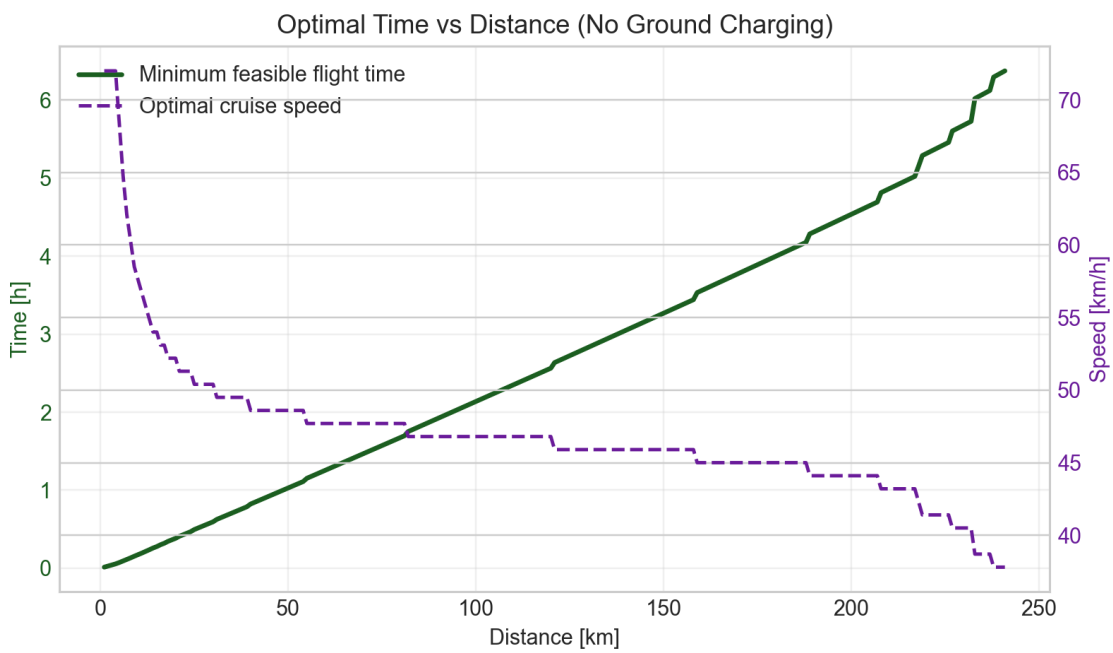

Figure 5: Time vs distance envelope without ground charging. For each distance, speed and departure time are optimized to minimize total flight time.

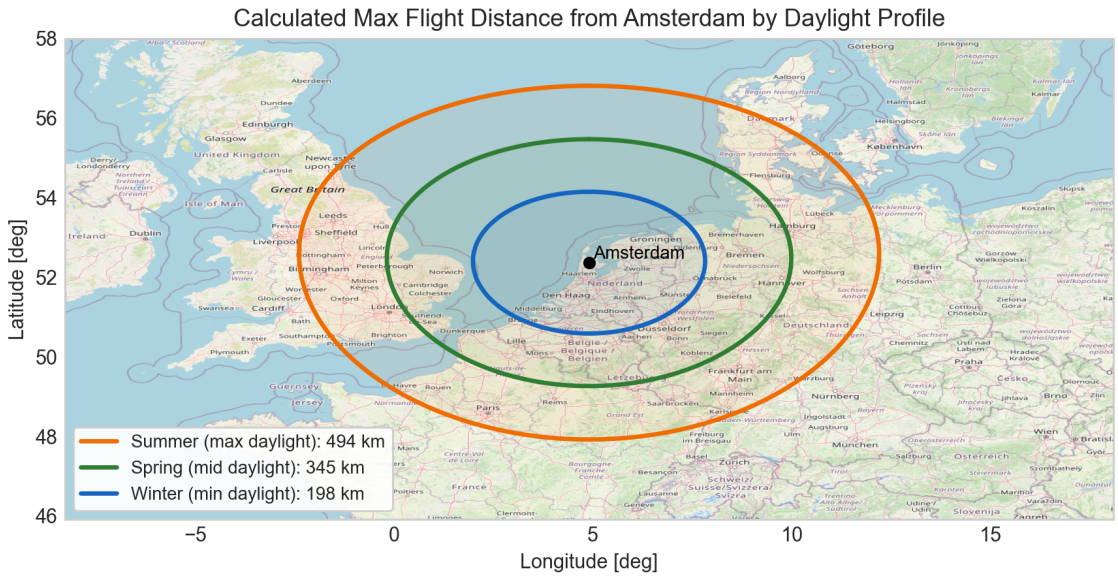

Figure 6: Range circles centered on Amsterdam using the calculated maximum distance for winter (min), spring (mid), and summer (max) daylight profiles.

Sample optimal points (no ground charging, battery starts at 100% SOC):

- 10 km: 0.17 h, optimal speed 57.6 km/h, optimal start 11.33h
- 25 km: 0.50 h, optimal speed 50.4 km/h, optimal start 11.00h
- 50 km: 1.03 h, optimal speed 48.6 km/h, optimal start 11.17h
- 75 km: 1.57 h, optimal speed 47.7 km/h, optimal start 11.00h
- 100 km: 2.14 h, optimal speed 46.8 km/h, optimal start 10.50h
- 150 km: 3.27 h, optimal speed 45.9 km/h, optimal start 10.17h
- 200 km: 4.54 h, optimal speed 44.1 km/h, optimal start 9.67h

Daylight profile assumptions and max-distance results:

- Winter (min daylight): daylight 7.75 h, peak irradiance 550 W/m^2, max distance 198.1 km (best speed 31.5 km/h, best start 9.50h).
- Spring (mid daylight): daylight 12.00 h, peak irradiance 850 W/m^2, max distance 345.0 km (best speed 36.0 km/h, best start 8.00h).
- Summer (max daylight): daylight 16.75 h, peak irradiance 1000 W/m^2, max distance 494.3 km (best speed 37.8 km/h, best start 7.17h).

4. Battery Feasibility for Your Battery Bay

- 30-minute reserve target: 15.16 Wh (minimum 1365 mAh at 3S nominal).
- 20-minute reserve target: 10.10 Wh.
- Theoretical max energy in bay (450–550 Wh/L): 14.31 Wh to 17.49 Wh.
- Daylight simulation minimum SOC: 0.0%, final SOC: 0.0%, first empty time: 8.33h.

Researched battery options

Pack	mAh	Wh	g	D x W x H (mm)	Strict fit	Meets 30-min reserve
Thunder Power Pro Lite V2 3S 500mAh 70C	500	5.55	49	53x30x13	Yes	No
Thunder Power Pro Lite V2 3S 750mAh 70C	750	8.32	69	55x31x18	No	No
Tattu R-Line 3S 650mAh 75C	650	7.21	70	58x31x16	No	No
Tattu R-Line 3S 850mAh 75C	850	9.43	91	58x30x22	No	No

Strict-fit options found: Thunder Power Pro Lite V2 3S 500mAh 70C

If depth can extend to about 58 mm: Thunder Power Pro Lite V2 3S 500mAh 70C, Thunder Power Pro Lite V2 3S 750mAh 70C, Tattu R-Line 3S 650mAh 75C

5. Proper MPPT Controller Selection (Module-level, not chip-only)

Using your 21-cell panel as one series string:

$$V_{MP,array} \approx 21 \cdot 0.55 = 11.55 \text{ V}, \quad V_{OC,array} \approx 21 \cdot 0.64 = 13.44 \text{ V}$$

This panel voltage is below robust buck-MPPT headroom for charging a 3S Li-ion pack at 12.6 V. For buck topologies, practical cell count is about 25 (minimum) to 28 (recommended margin). Therefore, with 21 cells, a **boost or buck-boost MPPT** is the correct architecture.

Selected converter for this build

Selected MPPT/buck-boost charger: TI BQ25798EVM

- Topology: Buck-boost charger with MPPT support
- Charge profile: Configure for 3S Li-ion (12.6V CV) through I2C
- Key specs used in this design: 3.6-24V input, 1-4 cell battery support, up to 5A charging current
- Fit with 21-cell + 3S architecture: Yes; buck-boost handles panel voltage below and above battery charge voltage

Required supporting components around MPPT

- PV fuse near panel positive lead, sized around 1.25x panel Isc branch current.
- Dedicated MPPT-to-battery branch fuse (about 10A).
- Main battery fuse for propulsion branch (about 40A with your ESC/motor class).
- XT60 (or equivalent) connectors and 16–14 AWG wire on high-current paths.

6. Wiring Diagram A: Power Architecture

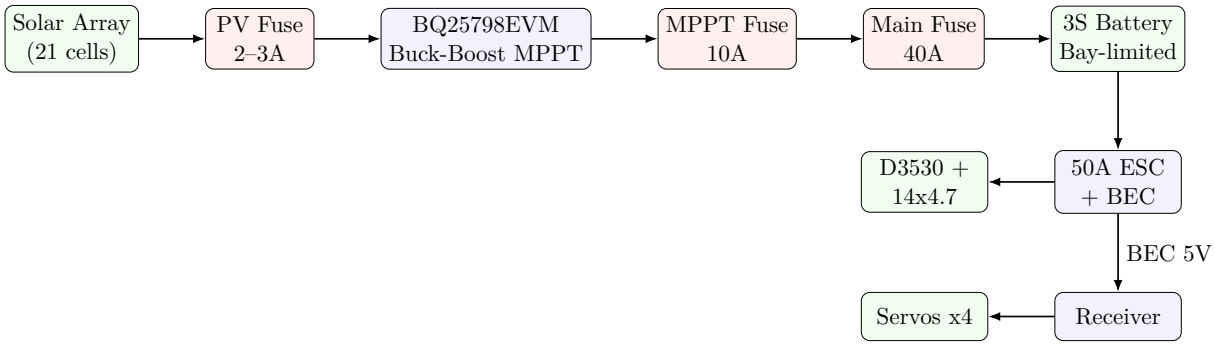

Figure 7: Recommended power wiring with dedicated MPPT branch protection.

7. Wiring Diagram B: Signal and Channel Mapping

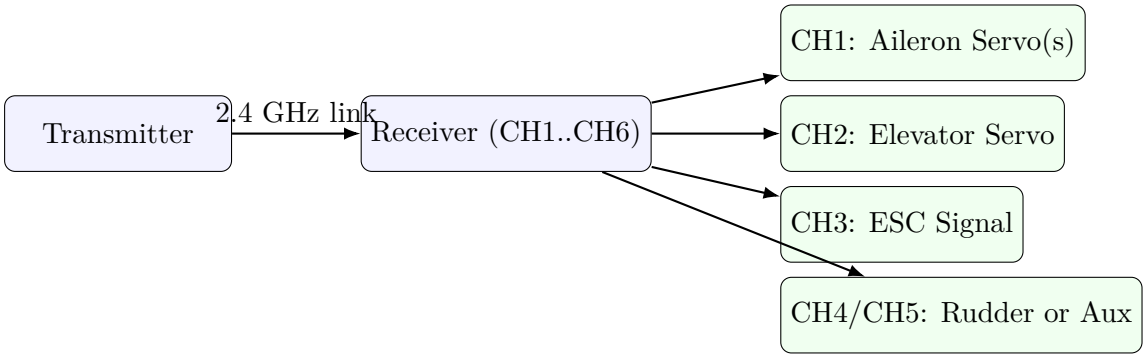

Figure 8: Signal wiring map. Verify channel assignments in transmitter before motor power-up.

8. Pin-level Wiring (Exact Connector Mapping)

8.1 Shared power-bus rule

Battery+, MPPT output+, and ESC+ are intentionally on the same positive DC bus node (through their branch fuses). Continuity between those + points is expected. A fault is only + to - continuity.

8.2 Receiver and ESC 3-wire lead pin mapping

Wire color (typical)	Pin function	Connect to receiver pin row	Notes
White/Yellow/Orange	Signal (S)	CH3 signal pin	ESC throttle command line
Red	+5V BEC output	CH3 + pin (receiver power rail)	Powers receiver and servos if ESC has BEC
Black/Brown	Ground (GND)	CH3 - pin	Must share common ground with all RC signal devices

8.3 Servo connector mapping

Receiver channel	Control surface	Pin order on receiver	Typical wire colors
CH1	Aileron	S / + / -	Orange or white / red / brown-black
CH2	Elevator	S / + / -	Orange or white / red / brown-black
CH4 (optional)	Rudder	S / + / -	Orange or white / red / brown-black
CH5/CH6 (optional)	Aux, flaps, telemetry trigger	S / + / -	Orange or white / red / brown-black

8.4 MPPT and high-current wiring map

Connection	Wire/fuse recommendation	Purpose
Solar+ to MPPT IN+	PV fuse 2-3A near panel	Protect panel branch from short faults
MPPT OUT+ to battery bus+	10A fuse near battery bus	Protect charger branch and wiring
Battery bus+ to ESC+	40A main fuse near battery	Protect propulsion high-current branch
All negatives (solar/MPPT/battery/ESC)	Common ground return bus	Required for charger and RC signal reference

8.5 Pre-power checklist

1. Remove propeller before bench wiring and ESC setup.
2. Confirm no hard short between main bus + and - (ohmmeter should not read near-zero steady value).
3. Confirm expected continuity between battery+, MPPT+, and ESC+ bus points.
4. Power receiver/ESC first from battery only, verify channel directions.
5. Connect solar and MPPT last, then confirm charging current with a wattmeter.

8.6 Final companion-power wiring (exact implementation)

- Battery bus + -> branch fuse 2A -> D36V50F5 VIN.
- D36V50F5 VOUT 5V -> Pi Zero 2 W 5V (power input).
- Battery bus + -> branch fuse 3A -> D30V33MAL VIN.
- Set D30V33MAL VOUT = 4.0V before connecting modem.
- D30V33MAL VOUT 4.0V -> A7670 VBAT.
- A7670 GND, Pi GND, FC GND, battery - all tied together.
- SpeedyBee F405 UART TX/RX <-> Pi UART RX/TX (crossed), same GND.
- Pi USB/UART (per modem interface choice) -> A7670 data link.

9. Wiring Diagram C: Pin-level Connector Orientation

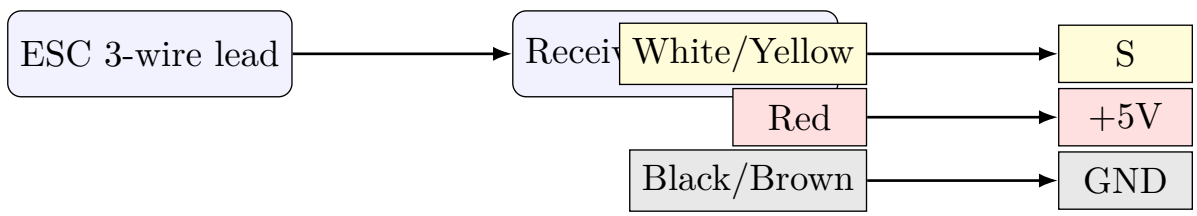

Figure 9: ESC lead to receiver CH3 mapping. Keep signal, +5V and GND aligned in the same row orientation.

10. LW-PLA and Carbon Tube Build Notes

- LW-PLA should be printed hot enough for foaming expansion; tune flow down after expansion to reduce mass.
- Carbon tubes should carry main bending loads in wing spar caps or spar tube; printed parts should primarily transfer loads into tubes.
- Reinforce wing root and motor mount interfaces with local carbon doublers or thicker load-spreading printed inserts.

11. Sources

1. Solar cell electrical reference (CSE125P-6BB): `https://solarinnova.net/en/product/cell-cse125p-6bb/`
2. TI BQ25798 product page (buck-boost charger with MPPT support): `https://www.ti.com/product/BQ25798`
3. TI BQ25798 datasheet: `https://www.ti.com/lit/gpn/bq25798`
4. TI BQ25798EVM user guide: `https://www.ti.com/lit/ug/sluucb5c/sluucb5c.pdf`

5. TI BQ24650 datasheet: https://www.ti.com/lit/gpn/bq24650
6. TI BQ24650 EVM user guide: https://www.ti.com/lit/ug/sluu444a/sluu444a.pdf
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18. OpenStreetMap tile service: https://tile.openstreetmap.org/